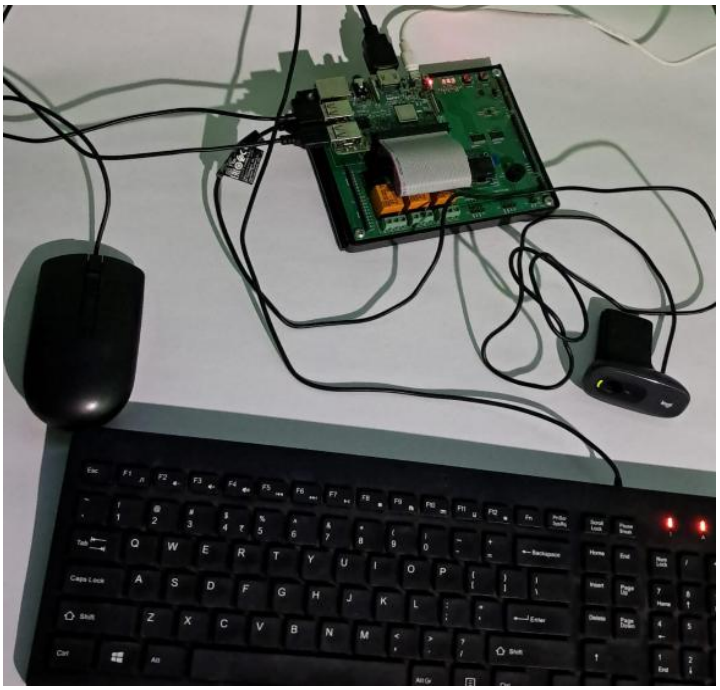
Project # 4

Gestures recognition on Raspberry pi

Student Team:

Aathikesavan B (III ECE) , **Manju Shree N B** & **Kavinesan B** (II ECE)

Working Model & Test Result



Description:

This project focuses on developing a **hand gesture recognition system using a Raspberry Pi** to enable natural and touchless human–computer interaction. A camera connected to the Raspberry Pi captures real-time images or video of hand movements, which are processed using computer vision and machine learning techniques. The system detects and classifies different hand gestures by analyzing features such as hand shape, motion, and finger positions. Lightweight algorithms or models such as **OpenCV-based image processing or CNN-based classifiers** are used so that the recognition can run efficiently on the Raspberry Pi platform.

The recognized gestures are then mapped to specific actions such as **controlling electronic devices, navigating a user interface, or operating a robot**, making the system useful for smart home control, assistive technologies, and interactive systems. By implementing the solution on Raspberry Pi, the project demonstrates a **low-cost, portable, and embedded AI system** capable of real-time gesture interpretation. This project integrates **embedded systems, computer vision, and edge AI**, providing a practical approach for intuitive and contactless device control.